

Lecture 1: Course Overview & Expectations

Rebecka Jörnsten, Mathematical Sciences

MSA220/MVE441 Statistical Learning for Big Data

23rd March 2026



CHALMERS
UNIVERSITY OF TECHNOLOGY



UNIVERSITY OF GOTHENBURG

Course Overview & Expectations

Who's involved

Rebecka Jörnsten

jornsten@chalmers.se

lectures and course coordination, examiner

Teaching Assistants



Isac Boström

isac.bostrom@chalmers.se



Lotta Eriksson

lottaer@chalmers.se

We will, each of us, hold office hours over zoom. Check Canvas for hours.

Who am I?

In my group we are 3 PhD students (and I co-supervise 4 more). We are always looking for master students!!!

- ▶ **SSF - CNSx3: organs-on-chips and rapid therapeutic development.** Using Q-learning and digital twins, we investigate how to come up with new therapeutic strategies against brain cancer.
- ▶ **DDLS - glioblastoma and cell tracking:** We combine NNs and statistical modeling to explain how brain cancer invades tissue and how treatments change the dynamics of cell-state transitions (branching, perivascular, round, locomotion).
- ▶ **Understanding NN through the lense of adaptive kernels:** My PhD student Vincent and I work on NN training and regularization and recast the NNs as kernel learners that are more transparent.
- ▶ **KAW consortium: Data integration in high-dimensional biology.** We investigate invasive routes of glioblastoma aiming to find the genetic mechanisms that determine why some tumors are invasive and some grow as bulk.

There's 135 of you and 3 of us...

Before you send an email/write a message

- ▶ Check on Canvas (Modules, Course PM, Syllabus, ...)
- ▶ Review the lecture slides
- ▶ If it's a content question: Think a little bit about the problem first and then **post in Discussions on Canvas**.

A course in three parts

1. Lectures
2. Projects
3. Take-home exam

- ▶ Data representations/Dimension reduction

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- ▶ Uncertainty quantification and ... some extras if we have time

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- ▶ We will focus on well-understood and interpretable methods and their modifications for big data sets.
- ▶ This course focuses on the statistics and not on the logistics of data.
- ▶ We can't cover everything BUT I hope that you will leave the class knowing how to approach using new methods, be risk aware, know how to properly validate and compare methods and feel comfortable about understanding pros and cons of types of methods

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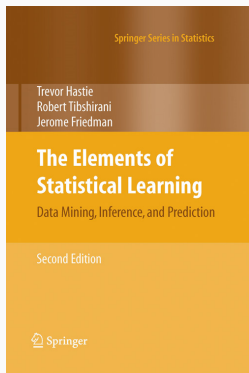
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- ▶ If you have taken an ML course: a bit more theory, analyzing assumptions, critical thinking about methods
- ▶ If you are a math/stat student: the ins and outs of applied statistics
- ▶ GOAL: that you leave this prepared to look for "red flags" and "alerts" in machine learning and data analysis, aware of main principles of statistical and machine learning methods, understanding limitations and assumptions and that you leave eager to learn more!



Hastie, T, Tibshirani, R, and Friedman, J (2009)
The Elements of Statistical Learning: Data Mining, Inference, and Prediction. 2nd ed. New York: Springer Science+Business Media, LLC

- ▶ Covers a lot of statistical methods
- ▶ Freely available online
- ▶ Balanced presentation of theory and application
- ▶ Not always very detailed. Other suggestions on course website.
- ▶ We will also use journal papers for some parts of the course

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- ▶ **Purpose:**
 - ▶ Hands-on experience in data-analysis
 - ▶ Further exploration of course topics
 - ▶ Practice how to present statistical results
 - ▶ Last but not least, preparation for the take-home exam

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 - ▶ The class will be divided into 3 parallel sessions for presentation day.

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- ▶ *Insufficient submissions may need to be reworked*
- ▶ Guidelines will be published on Canvas
- ▶ If possible, **please work and submit in pairs or up to a group of 5**

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- ▶ Use the programming language that you are most comfortable with.
- ▶ Don't divide the tasks between members of the group without discussion since that means you get a spotty prep for the exam...

- ▶ **Take-home exam**
- ▶ **Structure:** A couple of (carefully selected) data analysis tasks that need to be answered in form of individual reports.
- ▶ Exam will be posted on 29th of May
- ▶ Hard deadline on 5th June!!!
- ▶ We will talk more about the exam toward the end of the semester and I usually spend a proportion of the last lecture to go through it with you guys.

Can I use AI?

- ▶ Sure! However - if you are new to coding you will find that AI prompting might give you unnecessarily lengthy codes that is difficult to debug
- ▶ Also - if you are new to the methods we use, you might not spot that the AI rendered code using variants of a method that you did not intend to use. Example: tuning parameters added or default settings that are not what you intended to use - this might produce results that are in conflict with the theory that we have gone through in class. It is *your* responsibility to catch this! In the take-home exam, contradicting your own results due to such mistakes is heavily penalized.
- ▶ But - AI can save you a lot of time. Just make sure you know what the rendered code actually does.